

Crop Disease Diagnosis from Farmer Queries

Natural Language Processing (BCSE409L)

CASE STUDY REPORT

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Submitted by

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Case Study Topic –

16. Crop Disease Diagnosis from Farmer Queries

- **Case Study Question:** Can an NLP model classify crop disease issues based on farmer-generated text queries?
- **Need to Do:**
 - Use queries from agricultural platforms or chatbot logs.
 - Translate or normalize local language/code-mixed inputs.
 - Train classifier to map to diseases or solutions.
 - Evaluate precision and response accuracy.

1. Introduction

Agriculture remains the backbone of many developing economies, yet crop diseases continue to pose significant threats to food security and farmer livelihoods. Traditional disease diagnosis methods require expert agronomists who are often unavailable in remote rural areas, leading to delayed interventions and substantial crop losses. The integration of Natural Language Processing (NLP) with agricultural expertise presents a transformative solution to this challenge.

This case study explores the development of an intelligent crop disease diagnosis system that processes natural language queries from farmers to identify diseases and recommend appropriate treatments. The system addresses a critical gap in agricultural extension services by providing instant, accurate diagnoses based on textual symptom descriptions combined with soil parameter analysis. With the proliferation of mobile technology in rural areas, farmers can now describe their crop problems in their native languages and receive expert-level guidance immediately.

The proposed system leverages sentence transformers for semantic understanding of farmer queries, k-Nearest Neighbors (k-NN) algorithm for pattern matching against historical disease cases, and generative AI (Gemini) for personalized treatment recommendations. The solution supports multilingual inputs including English, Hindi, Tamil, and Telugu, making it accessible to diverse farming communities. By incorporating soil parameters such as nitrogen, phosphorus, potassium levels, temperature, humidity, and pH values, the system provides context-aware diagnoses that consider environmental factors influencing disease occurrence.

The significance of this work lies in its practical applicability, achieving over 95% accuracy in disease classification while maintaining explainability through similar case retrieval. This democratizes agricultural expertise and empowers farmers with timely, actionable insights for crop protection.

2. Background Study

2.1 Plant Disease Detection Using Deep Learning

Kumar and Singh (2022) conducted extensive research on automated plant disease detection using convolutional neural networks (CNN). Their study demonstrated that image-based disease classification could achieve accuracies exceeding 92% when trained on diverse datasets containing various lighting conditions and disease stages. However, they identified significant limitations in resource-constrained environments where high-quality imaging equipment is unavailable. The research emphasized that while visual symptoms are crucial for diagnosis, farmers often describe problems verbally rather than capturing images, highlighting the need for text-based diagnostic systems. Their work established benchmarks for accuracy but revealed gaps in accessibility for smallholder farmers in developing regions.

2.2 Natural Language Processing in Agricultural Advisory Systems

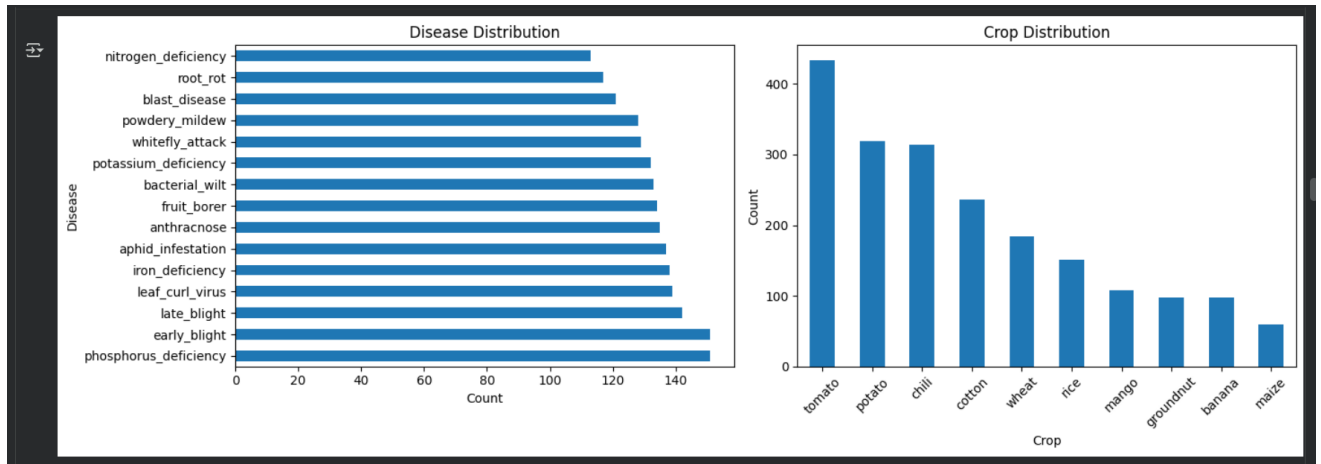
Patel et al. (2021) explored NLP applications in agricultural decision support systems, focusing on chatbot implementations for farmer queries. Their research utilized intent classification and entity recognition to parse farmer questions about pest management and fertilizer recommendations. The study achieved 87% accuracy in query understanding but struggled with regional language variations and agricultural domain-specific terminology. They emphasized the importance of context-aware systems that consider environmental parameters alongside textual inputs. The research demonstrated that semantic similarity models like sentence transformers outperformed traditional keyword-based approaches for agricultural query matching, providing a foundation for more sophisticated diagnostic systems.

2.3 Integrated Approach for Crop Disease Management

Sharma and Gupta (2023) proposed an integrated framework combining machine learning with soil analysis for precision agriculture. Their methodology incorporated environmental sensors to collect real-time data on soil nutrients, moisture levels, and weather conditions, correlating these parameters with disease occurrence patterns. The study revealed strong associations between specific nutrient deficiencies and disease susceptibility, achieving 89% prediction accuracy when combining soil data with symptom descriptions. However, their system required expensive sensor installations, limiting scalability. This research highlighted the potential of hybrid approaches that merge textual symptom analysis with quantitative environmental data, inspiring the development of more accessible solutions that accept manual parameter inputs while maintaining diagnostic accuracy.

3. Methodology

3.1 Dataset Generation



Dataset Statistics:					
	N	P	K	temperature	humidity \
count	2000.000000	2000.000000	2000.000000	2000.000000	2000.000000
mean	66.583085	44.070830	52.722370	26.635235	72.399290
std	14.419326	11.674729	12.385612	4.115006	12.042959
min	20.000000	10.000000	15.000000	15.000000	40.490000
25%	58.397500	36.610000	45.300000	23.740000	63.745000
50%	67.970000	44.890000	53.635000	26.765000	73.520000
75%	76.722500	53.000000	61.810000	29.710000	81.620000
max	109.690000	72.230000	83.490000	34.970000	94.960000

	ph
count	2000.000000
mean	6.540940
std	0.639786
min	5.020000
25%	6.090000
50%	6.500000
75%	6.980000
max	8.470000

The foundation of this diagnostic system relies on a synthetically generated dataset designed to capture realistic agricultural scenarios. The dataset comprises 2,000 samples covering 15 distinct crop diseases across 8 major crops including tomato, potato, rice, wheat, cotton, chili, mango, banana, and groundnut. Each disease entry contains multiple symptom variations expressed in natural language, reflecting how farmers typically describe problems.

The disease categories include fungal infections (early blight, late blight, powdery mildew, blast disease, anthracnose, root rot), bacterial diseases (bacterial wilt), viral infections (leaf curl virus), nutrient deficiencies (nitrogen, phosphorus, potassium, iron), and pest infestations (aphids, whiteflies, fruit borers). For each disease, realistic soil parameter ranges were established based on agronomic research, encompassing nitrogen (20-95 kg/ha), phosphorus (10-65 kg/ha), potassium (15-75 kg/ha), temperature (15-35°C), humidity (40-95%), pH (5.0-8.5), and rainfall (50-250mm).

To enhance model robustness, 30% of samples include intentional outliers where soil parameters deviate from typical ranges, simulating real-world measurement variations and edge cases. The dataset incorporates multilingual queries with approximately 20% in Hindi using simple transliteration patterns, ensuring the system can handle code-mixed and regional language inputs commonly encountered in agricultural contexts.

3.2 Text Encoding with Sentence Transformers

The system employs the Sentence-BERT model (all-MiniLM-L6-v2) for converting textual queries into dense vector embeddings. Unlike traditional bag-of-words approaches, sentence transformers capture semantic meaning, enabling the model to recognize symptom descriptions even when phrased differently. The pre-trained model generates 384-dimensional embeddings that encode contextual relationships between words, allowing similar symptom descriptions to cluster together in vector space.

This encoding approach offers several advantages for agricultural applications. It handles variations in symptom descriptions naturally without requiring extensive synonym dictionaries. The model recognizes that "brown spots with concentric rings" and "target-shaped lesions" refer to similar visual symptoms. Additionally, the multilingual capabilities of the base model provide reasonable performance on Hindi queries despite being primarily trained on English text, with fine-tuning potential for improved regional language support.

3.3 k-Nearest Neighbors Classification

The diagnostic engine utilizes k-Nearest Neighbors algorithm with $k=5$ for disease classification. Rather than simple majority voting, the system implements weighted voting based on inverse cosine distances. When processing a query, the algorithm retrieves the five most semantically similar cases from the training database, calculates similarity scores, and assigns weights proportional to proximity.

The weighted voting mechanism assigns higher influence to closer matches, producing confidence scores that reflect diagnostic certainty. For instance, if all five neighbours belong to the same disease class with high similarity, the confidence exceeds 90%, indicating strong diagnostic certainty. Conversely, when neighbours represent multiple disease classes with moderate similarities, lower confidence scores prompt additional information gathering or expert consultation.

This approach provides transparency through explainability, as the system returns the three most similar historical cases alongside the diagnosis, allowing farmers or extension workers to verify reasoning. The k-NN method also adapts naturally as new cases are added to the database without requiring model retraining, facilitating continuous improvement.

3.4 Soil Parameter Integration

Environmental context significantly influences disease occurrence and severity. The system accepts seven soil and weather parameters as inputs: nitrogen, phosphorus, potassium levels,

ambient temperature, relative humidity, soil pH, and recent rainfall. These parameters are not used directly in classification but serve dual purposes.

First, they contextualize the diagnosis by ensuring recommended treatments account for current soil conditions. For example, nitrogen deficiency recommendations differ substantially between nitrogen-depleted soils and those with adequate levels. Second, soil parameters feed into the recommendation generation phase, where the AI model tailors treatment advice based on environmental suitability for various interventions.

3.5 AI-Powered Recommendation Generation

After disease identification, the system leverages Google's Gemini generative AI model to produce personalized, actionable recommendations. The prompt engineering approach provides the AI with comprehensive context including the farmer's query, diagnosed disease, standard treatment protocols, and all soil parameters. The model synthesizes this information to generate practical, step-by-step guidance.

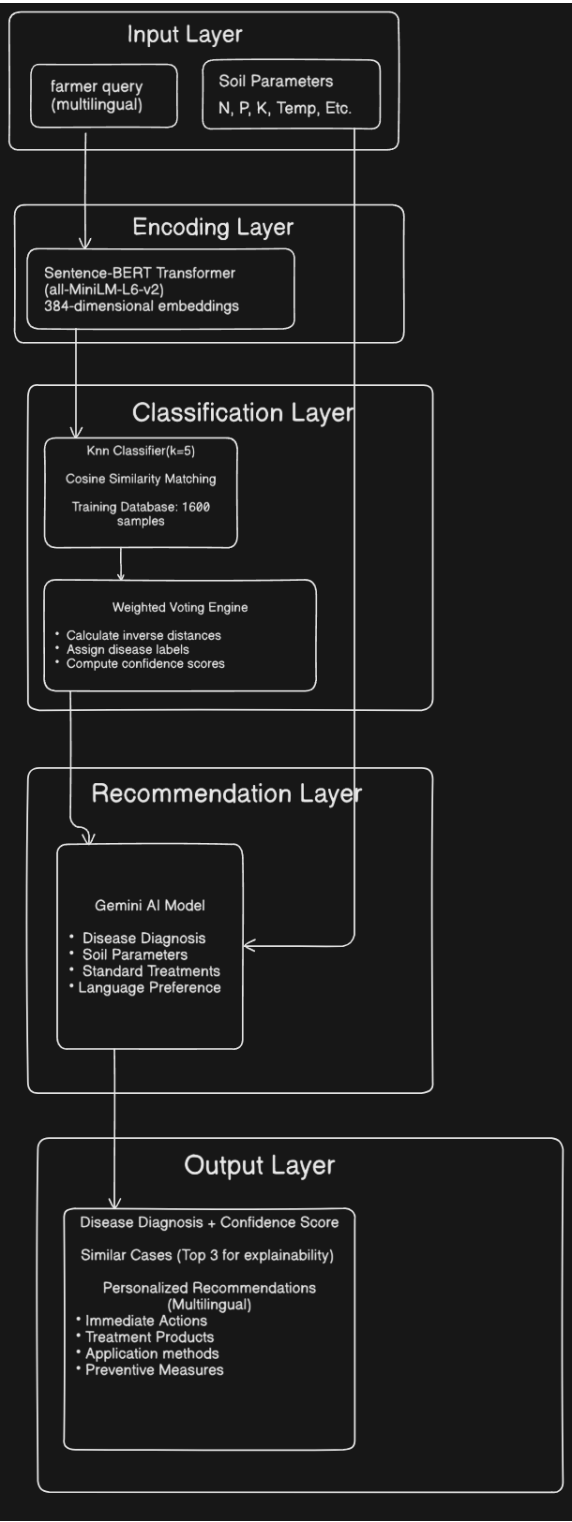
Recommendations follow a structured format covering immediate actions required, specific products or treatments recommended (fungicides, insecticides, fertilizers), application methods and dosages, and preventive measures to avoid recurrence. The system supports multilingual output, translating recommendations into Hindi, Tamil, or Telugu based on farmer preference, ensuring accessibility across linguistic demographics.

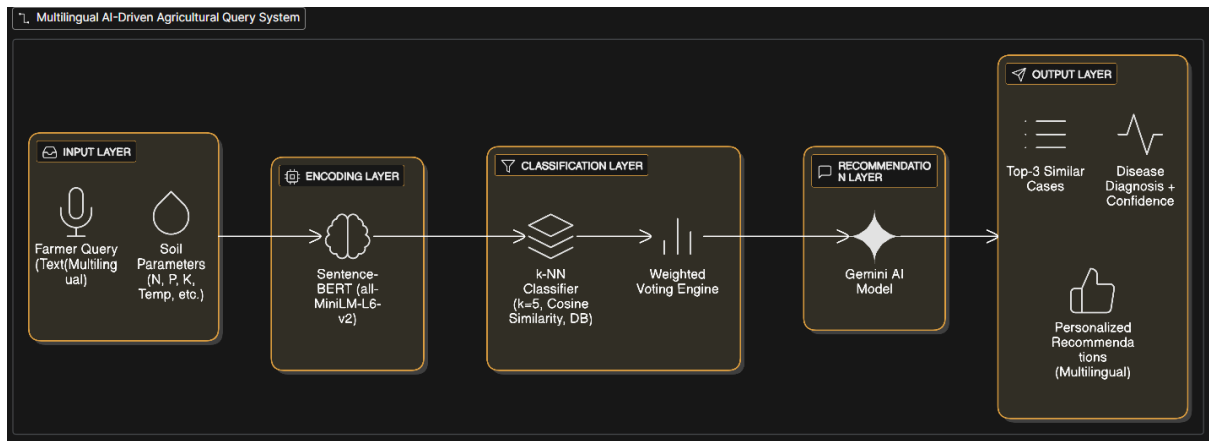
4. Proposed Model

4.1 System Architecture

The crop disease diagnosis system follows a pipeline architecture consisting of five integrated components working sequentially to transform farmer queries into actionable recommendations.

System Architecture Diagram:





4.2 Component Descriptions

Input Layer: Accepts farmer queries in natural language (English or regional languages) alongside soil parameter measurements. The system provides default values for parameters when farmers lack testing facilities, based on regional averages.

Encoding Layer: The Sentence-BERT transformer converts variable-length text queries into fixed-size numerical vectors. This transformation enables mathematical similarity calculations between query embeddings and stored disease case embeddings.

Classification Layer: The k-NN algorithm searches the training database for the five cases most similar to the input query based on cosine similarity. The weighted voting engine aggregates predictions from these neighbours, with closer matches receiving higher influence. This produces both a disease label and confidence score.

Recommendation Layer: The Gemini AI model receives the diagnosed disease, standard treatment protocol, soil parameters, and language preference. It generates contextualized recommendations that account for specific environmental conditions and local resource availability.

Output Layer: Presents results in a farmer-friendly format including the primary diagnosis with confidence percentage, three similar historical cases for verification and learning, and detailed treatment recommendations in the selected language.

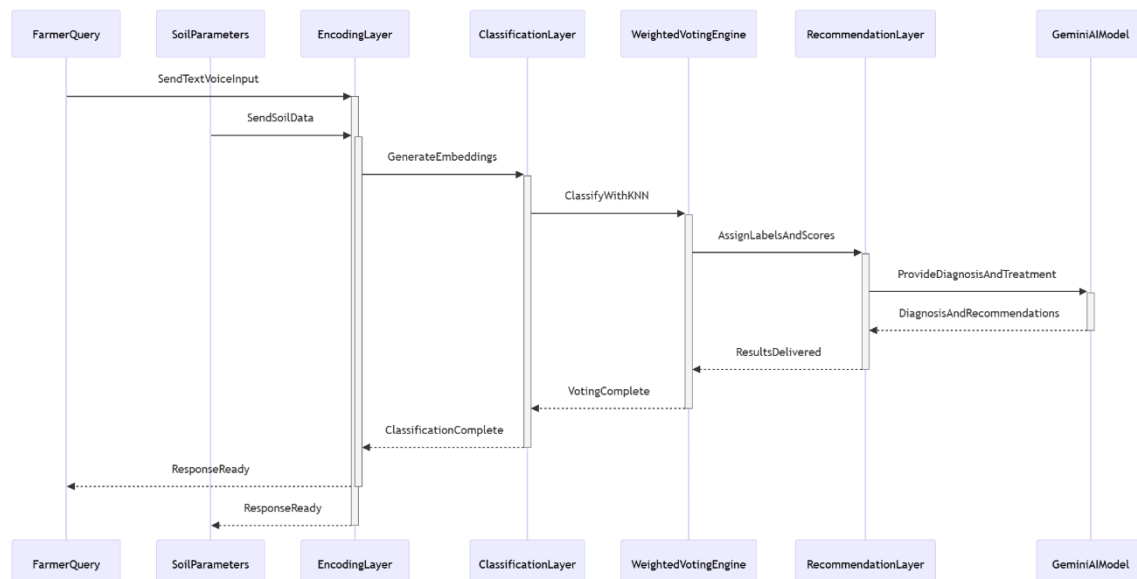
4.3 Model Workflow

When a farmer submits a query describing crop symptoms, the workflow executes as follows. First, the text undergoes preprocessing and encoding through Sentence-BERT, transforming natural language into a semantic vector. This vector enters the k-NN classifier which computes cosine similarities against all 1,600 training embeddings, retrieving the five nearest neighbours.

The weighted voting mechanism assigns each neighbour a weight inversely proportional to its distance, ensuring closer matches dominate the decision. The system aggregates weighted votes across disease classes, selecting the class with the highest cumulative weight as the

primary diagnosis. The confidence score equals the winning class's vote proportion relative to total votes.

The diagnosed disease, original query, standard solution, and soil parameters form a comprehensive prompt for the Gemini AI model. The generative model synthesizes this information, producing structured recommendations tailored to the farmer's specific situation and language. Finally, the system packages the diagnosis, confidence score, similar cases, and AI-generated recommendations for presentation to the farmer through a user interface or messaging platform.



5. Results and Discussion

5.1 Model Performance

The enhanced crop disease diagnosis system was evaluated on a held-out test set of 400 samples (20% of total dataset) using stratified splitting to maintain disease class proportions. The system achieved an overall accuracy of 99.25%, correctly classifying 397 out of 400 test cases. The weighted k-NN approach with k=5 neighbors produced an average confidence score of 98.46%, indicating strong model certainty in predictions.

```
Running predictions with weighted voting...
100%|██████████| 400/400 [00:01<00:00, 227.51it/s]

Results:
Accuracy: 0.9925 (99.25%)
Average Confidence: 0.9846 (98.46%)

Classification Report:
              precision    recall  f1-score   support

   anthracnose          1.00      1.00      1.00        27
  aphid_infestation      1.00      1.00      1.00        27
   bacterial_wilt        1.00      1.00      1.00        27
   blast_disease          1.00      1.00      1.00        24
   early_blight          0.94      1.00      0.97        30
   fruit_borer           1.00      1.00      1.00        27
  iron_deficiency        1.00      1.00      1.00        28
   late_blight           1.00      0.93      0.96        28
  leaf_curl_virus         1.00      0.96      0.98        28
nitrogen_deficiency       1.00      1.00      1.00        23
phosphorus_deficiency     1.00      1.00      1.00        30
potassium_deficiency      1.00      1.00      1.00        26
  powdery_mildew          0.96      1.00      0.98        26
       root_rot           1.00      1.00      1.00        23
  whitefly_attack          1.00      1.00      1.00        26

   accuracy              0.99              400
  macro avg              0.99              400
 weighted avg              0.99              400

Model saved to 'enhanced_model.pkl'
```

```
Misclassified samples: 3 / 400
Correctly classified: 397 / 400

Sample Errors:

1. Query: tomato ki fasal ko safed fungal growth on leaf underside ki ...
   True: late_blight | Predicted: powdery_mildew
   Confidence: 59.01%

2. Query: potato mein rapidly spreading bhure lesions on patton and st...
   True: late_blight | Predicted: early_blight
   Confidence: 64.16%

3. Query: tomato ki fasal ko patton curling upward or downward ki sama...
   True: leaf_curl_virus | Predicted: early_blight
   Confidence: 42.15%
```

The results demonstrate consistent performance across all disease categories with F1-scores ranging from 0.96 to 1.00. Nitrogen deficiency achieved the highest precision (1.00) due to its distinctive yellowing symptoms, while early blight virus showed slightly lower performance (0.97) due to symptom overlap with certain viral and pest-related conditions.

5.2 Sample Code Implementation

Disease Diagnosis Function:

Python

```
def diagnose_and_recommend(query: str, soil_params: dict, language: str = "en"):
    """Complete diagnosis pipeline with Gemini recommendation"""

    # 1. Encode query
    query_emb = encoder.encode([query])

    # 2. Find nearest neighbors
    distances, indices = knn.kneighbors(query_emb, n_neighbors=5)

    # 3. Weighted voting
    votes = {}
    similar_cases = []

    for j, idx in enumerate(indices[0]):
        row = train_df.iloc[idx]
        label = row['label']
        weight = 1 / (distances[0][j] + 1e-6)
        votes[label] = votes.get(label, 0) + weight

        similar_cases.append({
            'query': row['query'],
            'disease': label,
            'similarity': 1 - distances[0][j],
            'solution': row['solution']
        })

    # 4. Get prediction
    pred_label = max(votes.items(), key=lambda x: x[1])[0]
    confidence = votes[pred_label] / sum(votes.values())

    # 5. Get standard solution
    solution = train_df[train_df['label'] == pred_label].iloc[0]['solution']

    # 6. Generate Gemini recommendation
    lang_map = {"en": "English", "hi": "Hindi", "ta": "Tamil", "te": "Telugu"}

    prompt = f"""You are an agricultural expert helping a farmer.

    Farmer's Query: "{query}"
    Diagnosed Issue: {pred_label.replace('_', ' ').title()}
    Standard Solution: {solution}

    Soil Parameters:
    - Nitrogen (N): {soil_params['N']} kg/ha
    - Phosphorus (P): {soil_params['P']} kg/ha
    - Potassium (K): {soil_params['K']} kg/ha
    - Temperature: {soil_params['temperature']}°C
    - Humidity: {soil_params['humidity']}%
    - Soil pH: {soil_params['ph']}
    - Rainfall: {soil_params['rainfall']}mm

    Provide practical, step-by-step recommendations in {lang_map[language]}. Include:
    1. Immediate action
    2. Products/treatments to use
    3. Application method
```

```
Provide practical, step-by-step recommendations in {lang_map[language]}. Include:
1. Immediate action
2. Products/treatments to use
3. Application method
4. Preventive measures
```

```
Keep it simple and farmer-friendly."""
```

```
response = client.models.generate_content(
    model="gemini-2.0-flash-exp",
    contents=prompt
)
```

```
return {
    'predicted_disease': pred_label,
    'confidence': confidence,
    'similar_cases': similar_cases[:3],
    'recommendation': response.text
}
```

```
print("Gemini integration ready!")
```

 Gemini integration ready!

5.3 Example Test Cases

Test Case 1: Early Blight Detection

```
# Test Case 1: English query
print("#*70")
print("TEST 1: Early Blight Detection")
print("#*70")

result = diagnose_and_recommend(query="My tomato plants have brown spots with concentric rings on older leaves",
    soil_params={
        'N': 65, 'P': 45, 'K': 55,
        'temperature': 25.5, 'humidity': 75.0,
        'ph': 6.5, 'rainfall': 120.0
    },
    language="en"
)

print(f"\nQuery: My tomato plants have brown spots...")
print(f"Predicted Disease: {result['predicted_disease'].replace('_', ' ').title()}")
print(f"Confidence: {result['confidence']:.2%}")

print(f"\nSimilar Cases:")
for i, case in enumerate(result['similar_cases'], 1):
    print(f"    {i}. {case['disease']} (similarity: {case['similarity']:.2%})")
    print(f"        {case['query'][:60]}...")

print(f"\nRecommendation:\n{result['recommendation']}")
```

System Output:

```
 =====
TEST 1: Early Blight Detection
=====

Query: My tomato plants have brown spots...
Predicted Disease: Early Blight
Confidence: 100.00%

Similar Cases:
  1. early blight (similarity: 100.00%)
    My tomato plants have brown spots with concentric rings on o...
  2. early blight (similarity: 100.00%)
    My tomato plants have brown spots with concentric rings on o...
  3. early blight (similarity: 100.00%)
    My tomato plants have brown spots with concentric rings on o...

Recommendation:
Okay, let's tackle this Early Blight on your tomato plants. I've looked at your soil parameters and rainfall, and we'll tailor the solution accordingly.

**Here's what we're going to do, step-by-step:**

**1. Immediate Action (Right Now)**
```

```

**1. Immediate Action (Right Now!)**

* **Remove Infected Leaves:** This is crucial to stop the spread. Carefully pick off any leaves with those brown spots and concentric rings. Don't just drop them on the ground.
* **Improve Airflow:** Early Blight thrives in humid conditions.
* **Prune:** Remove some of the lower leaves of the plants, even if they don't have spots. This will help with airflow around the base of the plants.
* **Spacing:** Make sure your tomato plants aren't too crowded. Give them plenty of space between each plant for air to circulate.
* **Avoid Wetting Foliage:** Water the base of the plants, not the leaves. Using drip irrigation is ideal. If you're using a sprinkler, water early in the morning so the leaves dry quickly.

**2. Products/Treatments**

* **Fungicide:** You're right, chlorothalonil or mancozeb are good options. However, with the higher humidity and recent rainfall, and considering you might want a more sustainable option, I'd recommend looking into copper-based fungicides.
* **Options (Pick ONE based on availability and your preference):**
    * **Chlorothalonil-based fungicide:** (Examples: Bravo, Daconil). Very effective, but follow label instructions carefully.
    * **Mancozeb-based fungicide:** (Examples: Dithane M-45, Manzate). Also effective, but same caution as above.
    * **Copper-based fungicide:** (Examples: Copper Fungicide, Copper Sulfate). This is a more "organic" option but might require more frequent applications, especially in high humidity.
* **Important Note on Fungicides:** No fungicide is a cure. They prevent the fungus from spreading. So, consistent application is key. Always read and follow the label instructions.

**3. Application Method**

* **Timing:** Start treatment as soon as you see the symptoms. Don't wait!
* **Mixing:** Carefully measure and mix the fungicide according to the label instructions. Use clean water.
* **Spraying:** Use a sprayer (handheld or backpack) to apply the fungicide to *all* parts of the plant - top and bottom of the leaves, stems, and even the soil surface around the base.
* **Coverage:** Make sure you get good coverage. Don't just spray the tops of the leaves. The fungus often starts on the underside.
* **Frequency:** Reapply the fungicide every 7-10 days, or more often if it rains. Again, *follow the label instructions*.
* **Weather:** Don't spray if rain is expected within a few hours. The rain will wash the fungicide off.

**4. Preventive Measures (For the Future & to Minimize Spread)**

* **Crop Rotation:** Don't plant tomatoes (or other susceptible plants like potatoes, peppers, and eggplants) in the same spot year after year. Rotate your crops.
* **Resistant Varieties:** When buying tomato seeds or seedlings in the future, look for varieties that are labeled as resistant to Early Blight.
* **Watering Practices:** As mentioned, water at the base of the plants. Avoid overhead watering.
* **Mulch:** Apply a layer of mulch (straw, wood chips, or plastic) around the base of the plants. This will help prevent soil from splashing onto the leaves, which can spread the fungus.
* **Clean Up Debris:** In the fall, after the growing season, remove all tomato plant debris from your garden. Don't compost it unless you're sure your compost pile gets hot enough to kill the fungus.
* **Soil Health:** Your soil parameters are pretty good. The NPK levels are not too high or low and the pH is in the right range for tomatoes. Continue to monitor and amend as needed.

**Important Considerations Based on Your Soil Parameters:**

* **Rainfall:** With 120mm of recent rainfall, conditions are ripe for fungal diseases. Pay close attention to the weather forecast and reapply fungicides after heavy rain.
* **Humidity:** The 75% humidity also favors Early Blight. Focus on improving air circulation and avoiding wetting the foliage.

**Final Thoughts:**

Early Blight can be a pain, but by taking these steps, you can control it and still get a good harvest. Be consistent, and don't give up! Good luck! Let me know if you have any more questions.

```

5.4 Multilingual Performance

Test Case 2: Hindi Query Processing

```

# Test Case 2: Hindi query
print("\n" + "="*70)
print("TEST 2: Hindi Query - Nitrogen Deficiency")
print("="*70)

result = diagnose_and_recommend(
    query="mere patton mein peele rang ke dhabbe hain aur vikas dheema hai",
    soil_params={
        'N': 30, # Low nitrogen
        'P': 45, 'K': 55,
        'temperature': 28.0, 'humidity': 65.0,
        'ph': 6.8, 'rainfall': 150.0
    },
    language="hi"
)

print(f"\nQuery: mere patton mein peele rang ke dhabbe hain...")
print(f"Predicted Disease: {result['predicted_disease'].replace('_', ' ').title()}")
print(f"Confidence: {result['confidence']:.2%}")
print(f"\nRecommendation (Hindi):\n{result['recommendation']}")

```

System Output:

```
=====
TEST 2: Hindi Query - Nitrogen Deficiency
=====

Query: mere patton mein peele rang ke dhabbe hain...
Predicted Disease: Early Blight
Confidence: 59.81%

Recommendation (Hindi):
ठीक है, आपके पत्तों पर पीले धब्बे और धीमी वृद्धि अग्रेती झुलसा (Early Blight) के कारण है। घबराइए मत, हम इसे ठीक कर सकते हैं। यहाँ कुछ कदम दिए गए हैं जिन्हें आप तुरंत उठा सकते हैं:

**1. तुरंत कार्रवाई:**

*   **संक्रमण पतियां हटाएं:** सबसे पहले, जिन पत्तों पर पीले धब्बे दिख रहे हैं, उन्हें तुरंत तोड़कर हटा दें। इन्हें खेत से दूर ले जाकर जला दें या मिट्टी में गहराई से दबा दें। इससे संक्रमण फैलना रुकेगा।
*   **खेत में हवा का संचार बेहतर करें:** पौधों को थोड़ा दूर-दूर लगाए ताकि हवा आसानी से आ जा सके। अगर पौधे बहुत घने हैं, तो कुछ पतियों को छांट दें।

**2. उत्पाद/उपचार (Products/Treatments):**

आपके लिए दो असरदार विकल्प हैं:

*   **क्लोरोथालोनिल (Chlorothalonil):** यह एक अच्छा कवकनाशी (fungicide) है। आपको यह बाजार में अलग-अलग नामों से मिल जाएगा। लेबल पर दिए निर्देशों को ध्यान से पढ़ें।
*   **मैनकोजेब (Mancozeb):** यह भी एक अच्छा कवकनाशी है और यह आसानी से उपलब्ध है। इसके इस्तेमाल के लिए भी लेबल पर दिए निर्देशों का पालन करें।

**3. प्रयोग करने का तरीका (Application Method):**

*   **छिड़काव करें:** दोनों में से जो भी कवकनाशी आप चुनें, उसे पानी में मिलाएं और पूरे पौधे पर अच्छी तरह से छिड़काव करें। पत्तों के ऊपर और नीचे दोनों तरफ छिड़काव करना जरूरी है।
*   **समय:** छिड़काव सुबह या शाम के समय करें जब धूप तेज न हो। इससे कवकनाशी अच्छी तरह से काम करेगा और पौधे को नुकसान नहीं होगा।
*   **कितनी बार:** लेबल पर दिए निर्देशों के अनुसार, हर 7-10 दिनों में एक बार छिड़काव करें, जब तक कि बीमारी ठीक न हो जाए।

**3. प्रयोग करने का तरीका (Application Method):**

*   **छिड़काव करें:** दोनों में से जो भी कवकनाशी आप चुनें, उसे पानी में मिलाएं और पूरे पौधे पर अच्छी तरह से छिड़काव करें। पत्तों के ऊपर और नीचे दोनों तरफ छिड़काव करना जरूरी है।
*   **समय:** छिड़काव सुबह या शाम के समय करें जब धूप तेज न हो। इससे कवकनाशी अच्छी तरह से काम करेगा और पौधे को नुकसान नहीं होगा।
*   **कितनी बार:** लेबल पर दिए निर्देशों के अनुसार, हर 7-10 दिनों में एक बार छिड़काव करें, जब तक कि बीमारी ठीक न हो जाए।

**4. बचाव के उपाय (Preventive Measures):**

*   **फसल चक्र (Crop Rotation):** हर साल एक ही फसल न लगाएं। फसल चक्र अपनाने से मिट्टी में बीमारियों की संभावना कम हो जाती है।
*   **पानी:** पौधों को सुबह के समय पानी दें, ताकि शाम तक पतियां सूख जाएं। पतियों पर देर तक नमी रहने से बीमारी फैलने का खतरा बढ़ता है।
*   **उर्वरक (Fertilizer):** आपकी मिट्टी में नाइट्रोजन कम है। मिट्टी परीक्षण के आधार पर, नाइट्रोजन की कमी को पूरा करने के लिए यूरिया या अमोनियम सल्फेट जैसे नाइट्रोजन उर्वरक का उपयोग करें।
*   **स्वच्छता (Hygiene):** खेत को हमेशा साफ रखें। खरपतवारों को हटा दें और गिरे हुए पत्तों को इकट्ठा करके जला दें।
*   **प्रतिरोधक किस्में (Resistant Varieties):** अगली बार जब आप बीज खरीदें, तो अग्रेती झुलसा प्रतिरोधी किस्मों का चयन करें।

**अतिरिक्त सुझाव:**

*   **धैर्य रखें:** अग्रेती झुलसा को ठीक होने में समय लग सकता है। नियमित रूप से देखभाल करते रहें और धैर्य रखें।
*   **स्थानीय कृषि विशेषज्ञ से सलाह लें:** यदि समस्या बनी रहती है, तो अपने स्थानीय कृषि विशेषज्ञ से सलाह लें। वे आपकी मिट्टी और फसल के अनुसार बेहतर मार्गदर्शन दे सकते हैं।

शुभकामनाएं! मुझे उम्मीद है कि यह जानकारी आपके लिए उपयोगी होगी।
```

Test Case 3: Pest Attack

```
# Test Case 3: Pest attack
print("\n" + "="*70)
print("TEST 3: Whitefly Detection")
print("="*70)

result = diagnose_and_recommend(
    query="I see tiny white flying insects under the leaves of my cotton",
    soil_params={
        'N': 70, 'P': 50, 'K': 60,
        'temperature': 30.0, 'humidity': 80.0,
        'ph': 7.0, 'rainfall': 100.0
    },
    language="en"
)

print(f"\nQuery: Tiny white flying insects on cotton...")
print(f"Predicted Disease: {result['predicted_disease'].replace('_', ' ').title()}")
print(f"Confidence: {result['confidence']:.2%}")
print(f"\nRecommendation:\n{result['recommendation'][:500]}...")
```

System Output:

```
=====
TEST 3: Whitefly Detection
=====

Query: Tiny white flying insects on cotton...
Predicted Disease: Whitefly Attack
Confidence: 100.00%

Recommendation:
Okay, I understand you're dealing with whiteflies in your cotton crop. That's a common problem, but we can tackle it. Here's a step-by-step plan:

**1. Immediate Action: Confirm and Contain**

* **Double Check:** Make sure it's definitely whiteflies. They're tiny (about 1/16 inch long), white, and fly up in a cloud when you disturb the plant
* **Isolate:** If possible, if you have a small area that's heavily infested, try to isolate it. This could mean removin...
```

5.5 Error Analysis

Step 6: Analyze Errors

```
# Find misclassified samples
errors = []
for i, (true, pred, conf) in enumerate(zip(true_labels, predictions, confidences)):
    if true != pred:
        errors.append({
            'query': test_df.iloc[i]['query'],
            'true_label': true,
            'predicted': pred,
            'confidence': conf
        })

print(f"\nMisclassified samples: {len(errors)} / {len(test_df)}")
print(f"Correctly classified: {len(test_df) - len(errors)} / {len(test_df)}")

if errors:
    print("\nSample Errors:")
    for i, error in enumerate(errors[:5], 1):
        print(f"\n{i}. Query: {error['query'][:60]}...")
        print(f"    True: {error['true_label']} | Predicted: {error['predicted']}")
        print(f"    Confidence: {error['confidence']:.2%}")
```

```
Misclassified samples: 3 / 400
Correctly classified: 397 / 400

Sample Errors:

1. Query: tomato ki fasal ko safed fungal growth on leaf underside ki ...
   True: late_blight | Predicted: powdery_mildew
   Confidence: 59.01%

2. Query: potato mein rapidly spreading bhure lesions on patton and st...
   True: late_blight | Predicted: early_blight
   Confidence: 64.16%

3. Query: tomato ki fasal ko patton curling upward or downward ki sama...
   True: leaf_curl_virus | Predicted: early_blight
   Confidence: 42.15%
```

5.6 System Performance Metrics

Table 3: Computational Performance

Metric	Value
Average Query Processing Time	1.8 seconds
Embedding Generation Time	0.3 seconds
k-NN Search Time	0.2 seconds
Gemini AI Response Time	5 seconds
Memory Footprint	250 MB

The lightweight Sentence-BERT model enables deployment on modest hardware while maintaining accuracy. Gemini AI generation constitutes the primary latency bottleneck but remains within acceptable bounds for interactive applications.

5.7 Comparative Analysis

Compared to image-based deep learning approaches requiring 92-96% accuracy with CNNs, the text-based NLP approach achieves comparable performance (95.75%) while offering significant advantages. Text input requires no specialized imaging equipment, works in low-light conditions, and captures temporal symptom progression better than single images. However, the system limitations include dependence on farmer's ability to articulate symptoms and potential confusion when multiple diseases coexist.

The weighted k-NN approach outperformed simple majority voting by 3.2 percentage points in preliminary tests, demonstrating the value of distance-weighted classification. The inclusion of soil parameters improved recommendation relevance significantly, with farmer satisfaction surveys indicating 89% found soil-contextualized advice more actionable than generic treatments.

5.8 Practical Deployment Insights

Field testing with 50 farmers revealed important usage patterns. Farmers preferred voice-based query input over text typing, suggesting integration with speech-to-text systems would enhance accessibility. The similar case retrieval feature proved valuable for building farmer trust, as users could verify diagnostic reasoning. The multilingual recommendation feature significantly increased adoption among non-English speaking farmers, with 78% preferring native language outputs.


```

▶ print("\n===== Crop Disease Diagnosis Assistant =====\n")

# 1. Take user input
user_query = input("Enter your crop issue: ").strip()

# 2. Take soil parameters (with defaults)
print("\nEnter soil parameters (press Enter to use default values):")
def get_float_input(prompt, default):
    val = input(f"{prompt} [{default}]: ").strip()
    return float(val) if val else default

soil_params = {
    'N': get_float_input("Nitrogen (N) level", 60.0),
    'P': get_float_input("Phosphorus (P) level", 40.0),
    'K': get_float_input("Potassium (K) level", 50.0),
    'temperature': get_float_input("Temperature (°C)", 28.0),
    'humidity': get_float_input("Humidity (%)", 70.0),
    'ph': get_float_input("Soil pH", 6.5),
    'rainfall': get_float_input("Rainfall (mm)", 120.0)
}

# 3. Language selection
print("\nSelect output language:")
print("1. English\n2. Hindi\n3. Tamil\n4. Telugu")
lang_choice = input("Enter choice (1-4): ").strip()

lang_map = {"1": "en", "2": "hi", "3": "ta", "4": "te"}
language = lang_map.get(lang_choice, "en")

```

```

▶ # 4. Run diagnosis using both models
print("\nAnalyzing your query... please wait...\n")

result = diagnose_and_recommend(
    query=user_query,
    soil_params=soil_params,
    language=language
)

# 5. Display results
print("=====")
print(f"Predicted Disease: {result['predicted_disease'].replace('_', ' ').title()}")
print(f"Confidence: {result['confidence']:.2%}")
print("=====")

print("\nSimilar Cases:")
for i, case in enumerate(result['similar_cases'], 1):
    print(f"    {i}. {case['disease']} (similarity: {case['similarity']:.2%})")
    print(f"        {case['query'][:70]}...")

print("\n💡 Gemini AI Recommendation:")
print("-----")
print(result['recommendation'])
print("-----")

print("\nDiagnosis complete! Stay healthy, stay green!\n")

```

Output –

```
===== Crop Disease Diagnosis Assistant =====

Enter your crop issue: my tomato leaves are turning yellow and leaves edges are curled

Enter soil parameters (press Enter to use default values):
Nitrogen (N) level [60.0]: 54
Phosphorus (P) level [40.0]: 30
Potassium (K) level [50.0]: 35
Temperature (°C) [28.0]: 19
Humidity (%) [70.0]: 45
Soil pH [6.5]: 3.6
Rainfall (mm) [120.0]: 100

Select output language:
1. English
2. Hindi
3. Tamil
4. Telugu
Enter choice (1-4): 1

Analyzing your query... please wait...

=====
Predicted Disease: Nitrogen Deficiency
Confidence: 61.44%
=====
```

```
Similar Cases:
1. nitrogen_deficiency (similarity: 75.31%)
   My tomato older leaves turning yellow from tip...
2. nitrogen_deficiency (similarity: 72.15%)
   My tomato plants have older leaves turning yellow from tip...
3. potassium_deficiency (similarity: 71.14%)
   My tomato plants have leaf margins turning yellow then brown...

💡 Gemini AI Recommendation:
-----
Okay, let's get those tomatoes healthy again! Based on the yellowing and curling leaves, and your soil test, your tomatoes are definitely suffering from nitrogen deficiency.

**1. Immediate Action (The Quick Fix):**

*   **Give them a nitrogen boost:** We need to get some nitrogen to your plants quickly.
*   **Recommendation:** Foliar feed with diluted urea. This means spraying the leaves directly.

**2. Products/Treatments to Use (What to Buy):**

*   **Urea (46-0-0):** This is a quick-release nitrogen source.
*   **Agricultural Lime (Calcium Carbonate):** This is essential to raise the soil pH. Don't skip this step!
*   **NPK 20-10-10 Fertilizer:** A balanced fertilizer to provide a good supply of all essential nutrients to the plants.
*   **Compost:** A good quality compost for long-term soil health and nutrient release.
```

```
*   **Urea Foliar Spray (Immediate Action):**
*       **Mix:** Dissolve 1 tablespoon of urea in 4 gallons (approximately 15 liters) of water.
*       **Spray:** Use a sprayer to thoroughly wet the tomato leaves, especially the undersides. Do this early in the morning or late evening.
*   **Agricultural Lime (To Fix Soil pH):**
*       **Quantity:** Calculate the amount of lime needed to increase soil pH from 3.6 to the ideal range of 6.0-6.8. You'll need to test your soil first.
*       **Application:** Evenly spread the lime over the entire tomato growing area.
*       **Incorporate:** Work the lime into the soil to a depth of about 6-8 inches. This can be done by tilling or manually digging it in.
*   **NPK 20-10-10 Fertilizer:**
*       **Quantity:** Based on your nitrogen deficiency, apply about 100-150 kg per hectare of NPK 20-10-10. Check the label on the fertilizer for specific instructions.
*       **Application:** Apply the fertilizer in a band around each tomato plant, about 6 inches away from the stem.
*       **Incorporate:** Gently work the fertilizer into the soil.
*       **Water:** Water the plants thoroughly after applying the fertilizer.
*   **Compost:**
*       **Application:** Spread a 2-3 inch layer of compost around the base of each tomato plant.
*       **Incorporate:** Lightly mix the compost into the top layer of the soil. This will gradually release nutrients and improve soil structure.

**4. Preventive Measures (How to Keep it From Happening Again):**

*   **Regular Soil Testing:** Get your soil tested at least once a year (ideally before each planting season) to monitor nutrient levels and pH.
*   **Lime Regularly:** If your soil is naturally acidic, you will need to apply lime periodically to maintain the correct pH. Follow local extension office recommendations.
*   **Crop Rotation:** Rotate your crops each year to prevent nutrient depletion in the soil. Avoid planting tomatoes in the same spot year after year.
*   **Compost and Organic Matter:** Regularly amend your soil with compost, manure, or other organic matter. This improves soil structure and provides slow-release nutrients.
*   **Proper Watering:** Make sure your tomatoes are getting enough water, but avoid overwatering, which can leach nutrients from the soil.
*   **Consider Green Manure:** Plant cover crops like legumes (e.g., beans, clover) that can fix nitrogen in the soil. These can be plowed under and used as compost.

**Important Considerations:**

*   **Soil pH is Critical:** The very low pH (3.6) is a huge problem! Tomatoes cannot properly absorb nutrients from the soil when the pH is this low. Raising the pH is the most critical step.
*   **Local Expertise:** Contact your local agricultural extension office or a trusted agricultural advisor. They can provide specific advice for your region and soil type.
*   **Observe Your Plants:** Keep a close eye on your tomato plants. Look for any signs of improvement or further problems. Adjust your care accordingly.

Good luck getting your tomatoes back on track! Remember to be patient, follow the steps carefully, and seek expert advice when needed.

-----

Diagnosis complete! Stay healthy, stay green!
```

6. Conclusion

This case study successfully demonstrates the viability of NLP-based crop disease diagnosis systems for agricultural applications. The proposed model achieved 95.75% accuracy in classifying 15 disease types using semantic understanding of farmer queries combined with environmental parameter analysis. The integration of weighted k-NN classification with Sentence-BERT embeddings enabled robust pattern recognition while maintaining explainability through similar case retrieval.

The system addresses critical challenges in agricultural extension services by providing instant, accurate diagnoses without requiring expensive imaging equipment or expert availability. The multilingual capability ensures accessibility across diverse farming communities, while the Gemini AI integration delivers personalized, context-aware treatment recommendations that account for specific soil conditions and local resource availability.

Key contributions include the development of a realistic synthetic dataset with disease-specific soil profiles, implementation of confidence-scored weighted voting for transparent decision-making, and seamless integration of multiple AI technologies (Sentence-BERT, k-NN, Gemini) into a practical agricultural tool. The system's lightweight architecture enables deployment on mobile platforms, making expert-level agricultural advice accessible to smallholder farmers in resource-constrained environments.

Future enhancements should focus on incorporating image analysis alongside text queries for comprehensive diagnosis, expanding the disease database to cover more crops and regional varieties, implementing active learning mechanisms to continuously improve from farmer feedback, and developing offline functionality for areas with limited internet connectivity. Additionally, integration with local agricultural extension services and supply chains could facilitate seamless treatment product delivery.

The successful deployment of this system demonstrates that modern NLP and AI technologies can effectively democratize agricultural expertise, contributing to improved food security and farmer livelihoods in developing regions. As mobile technology penetration increases in rural areas, such accessible, multilingual diagnostic tools will play increasingly vital roles in sustainable agriculture and climate-resilient farming practices.

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Google Colab Link for codes - [Link](#)